

INTER DEPARTMENTAL PROJECT REPORT

GSM BASED AGRICULTURAL MOTOR CONTROL

Submitted in Partial Fulfilment of the Requirements for the Degree of

Electronics and Communication Engineering

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CERTIFICATE

This is to certify that the Inter departmental Project Report entitled **“GSM BASED AGRICULTURAL MOTOR CONTROL”** that is being submitted by **Y.Sushma(191FA04059),D.Tejaswini(191FA04572),G.Goutham(191FA05025), M.LalithKrishna(201LA05001)** in partial fulfilment for the award of B.Tech degree in CSE & ECE to the Vignan’s Foundation for Science, Technology and Research, Deemed to be University, is a record of bonafide work carried out by them at **Department of CSE & ECE** under the supervision of Dr.Srilakshmi Uppalapati(CSE) ,Mr.Taj(ECE)

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1. ABSTRACT

This Project provides an excellent solution for the need not go to the farm to water the field. In this project, we are using Arduino, which is interfaced with a GSM modem to send the message about the motor for particular member. Here how it works: Moisture Sensor ,DHT-11,Rain Sensor,GSM Modem to the Microcontroller.

When the moisture sensor detect no water in the field the information immediately send to the particular member through GSM modem, automatically motor gets ON. When Moisture Sensor detect water in the field the information immediately send to the particular member through GSM modem, automatically Motor gets off. If there is heavy rain the water level is more in the field which is calculated by the rain sensor, that information goes to the Particular member, by that he can control the motor in the field through GSM modem.GSM module is nothing but the sim that is connected to the Arduino.

Through this we can easily improve our irrigation system.

2. INTRODUCTION

The idea behind this project is powered by the problems faced by the Indian Farmers. You might have noticed that in most of the cases the crop-fields are located far away from the farmer's residence. Farmers have to travel significant distances to go to their crop fields.

Irrigation is an essential part of farming. As we know, it needs to be done daily, and due to this fact, whatever may be the distance between farm and home, the farmer has to go to their crop fields every day to turn on the irrigation system and supply water to crops.

There are two cons of this daily travel to crop fields, waste of both time and money.

So, we felt like traveling is not a viable option when it can be done remotely. With GSM based agriculture motor control it's that simple; sitting at home, clicking a button on your smart-phone and controlling motors.

3. PROBLEM STATEMENT

This Project helps the farmers for better Irrigation Process. This Project main aim is to design an improved irrigation and a user-friendly system so that , it can help the farmer to regulate the schedule of watering crops.

4. REQUIREMENTS

4.1 SOFTWARE REQUIREMENTS

- Arduino IDE

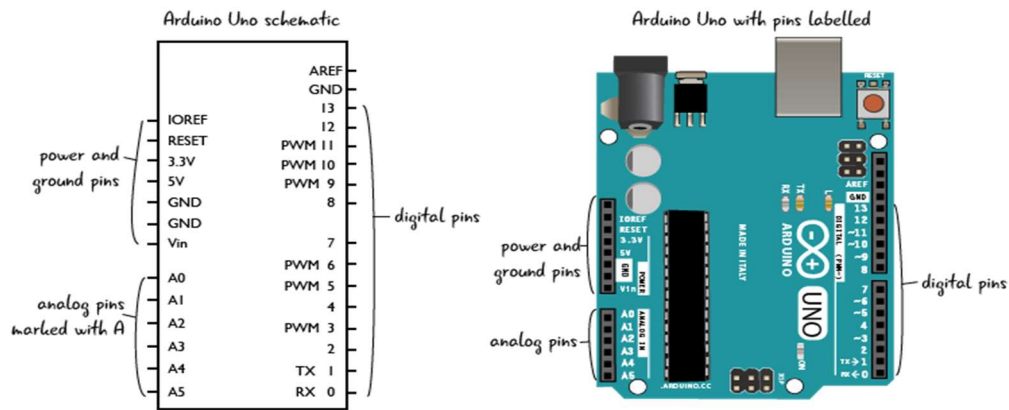
4.2 HARDWARE REQUIREMENTS

- Arduino UNO
- DHT-11
- Soil Moisture Sensor
- Rain Sensor
- GSM Module
- Jumper Wire
- DC Motor and Relay

5. Components

➤ Arduino UNO:

It is an open source microcontroller board based on the ATmega328P (datasheet) Consists of 14 digital input/output pins (of which 6 can be utilized as PWM outputs), 6 analog input pins, a 16 MHz crystal oscillator, a USB association, a power jack, an ICSP header, and a reset catch. It contains everything in built expected to help the microcontroller, basically associate it to a PC with a USB cable or power it with an AC-to-DC connector or battery to begin.

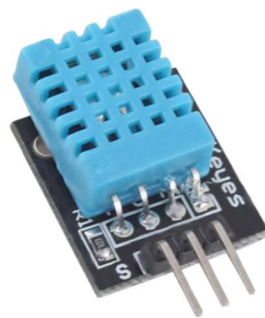


ARDUINO Specifications & Features :

Microcontroller	ATmega38P – 8 bit AVR family microcontroller
Operating Voltage	5V
Recommended Input Voltage	7-12V
Input Voltage Limits	6-20V
Analog Input Pins	6 (A0-A5)
Digital I/O Pins	14 (Out of which 6 provide PWM output)
DC Current on I/O Pins	40mA
DC Current on 3.3V Pin	50mA
Flash Memory	32 KB (0.5 KB is used for Bootloader)
SRAM	2kB
EEPROM	1kB

➤ DHT-11:

The DHT11 is a basic, ultra low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air, and spits out a digital signal on the data pin (no analog input pins needed). Its fairly simple to use, but requires careful timing to grab data. You can get new data from it once every 2 seconds, so when using the library from Adafruit, sensor readings can be up to 2 seconds old. Comes with a 4.7K or 10K resistor, which you will want to use as a pullup from the data pin to VCC.



DHT-11

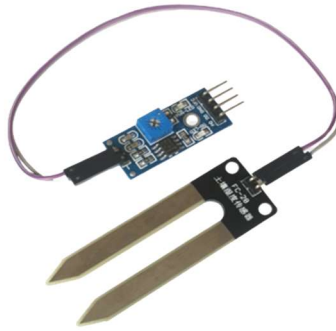
DHT-11 Specifications:

- 3 to 5V power and I/O
- 2.5mA max current use during conversion (while requesting data)
- Good for 20-80% humidity readings with 5% accuracy
- Good for 0-50 °C temperature readings ± 2 °C accuracy
- No more than 1 Hz sampling rate (once every second)
- Body size 15.5mm x 12mm x 5.5mm
- 4 pins with 0.1" spacing
- RoHS compliant

➤ Soil Moisture Sensor:

The soil moisture sensor is one kind of sensor used to gauge the volumetric content of water within the soil. As the straight gravimetric dimension of soil moisture needs eliminating, drying, as well as sample weighting. These sensors measure the volumetric water content not directly with the help of some other rules of soil like

dielectric constant, electrical resistance, otherwise interaction with neutrons, and replacement of the moisture content.



SOIL MOISTURE SENSOR

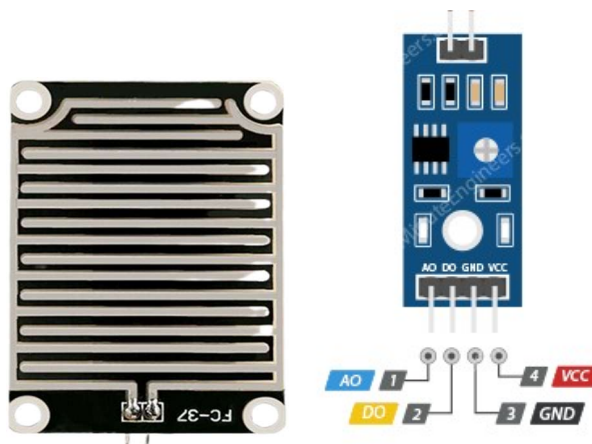
This Sensor consists of 4 pins:

- VCC pin is used for power
- A0 pin is an analog output
- D0 pin is a digital output
- GND pin is a Ground

Soil moisture Specifications:

- The required voltage for working is 5V.
- The required current for working is <20mA
- The required temperature of working of this sensor is 10°C to 30°C

➤ **Rain Sensor:**



Rain Sensor

Rain Sensor module

This Sensor consists of 4 pins:

- VCC pin is used for power
- A0 pin is an analog output
- D0 pin is a digital output
- GND pin is a Ground

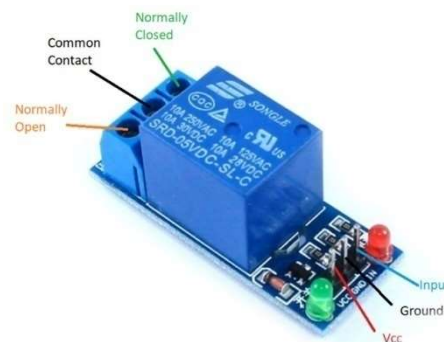
Soil moisture Specifications:

- The required voltage for working is 3.3v to 5V.
- The required current for working is 15mA
- The sensing pad size is 5cm x 4 cm with a nickel plate on one face.

➤ DC Motor and Relay Module:



DC Motor



Relay Module

A **power relay module** is an electrical switch that is operated by an electromagnet. The electromagnet is activated by a separate low-power signal from a micro controller. When activated, the electromagnet pulls to either open or close an electrical circuit.

DC water pump is a machine that transports liquid or pressurizes liquid. When the water pump is working, the coil and commutator rotate, but the magnetic steel and carbon brushes do not rotate. The alternating current direction of the coil is changed by the commutator and brushes that rotate with the motor.

-A **DC motor** is any of a class of rotary electrical motors that converts direct current (DC) electrical energy into mechanical energy. The most common types rely on the forces produced by magnetic fields. Nearly all types of DC motors have some internal mechanism, either electromechanical or electronic, to periodically change the direction of current in part of the motor.

➤ GSM module:



GSM Module

A customized **Global System for Mobile communication (GSM) module** is designed for wireless radiation monitoring through Short Messaging Service (SMS). This module is able to receive serial data from radiation monitoring devices such as survey meter or area monitor and transmit the data as text SMS to a host server.

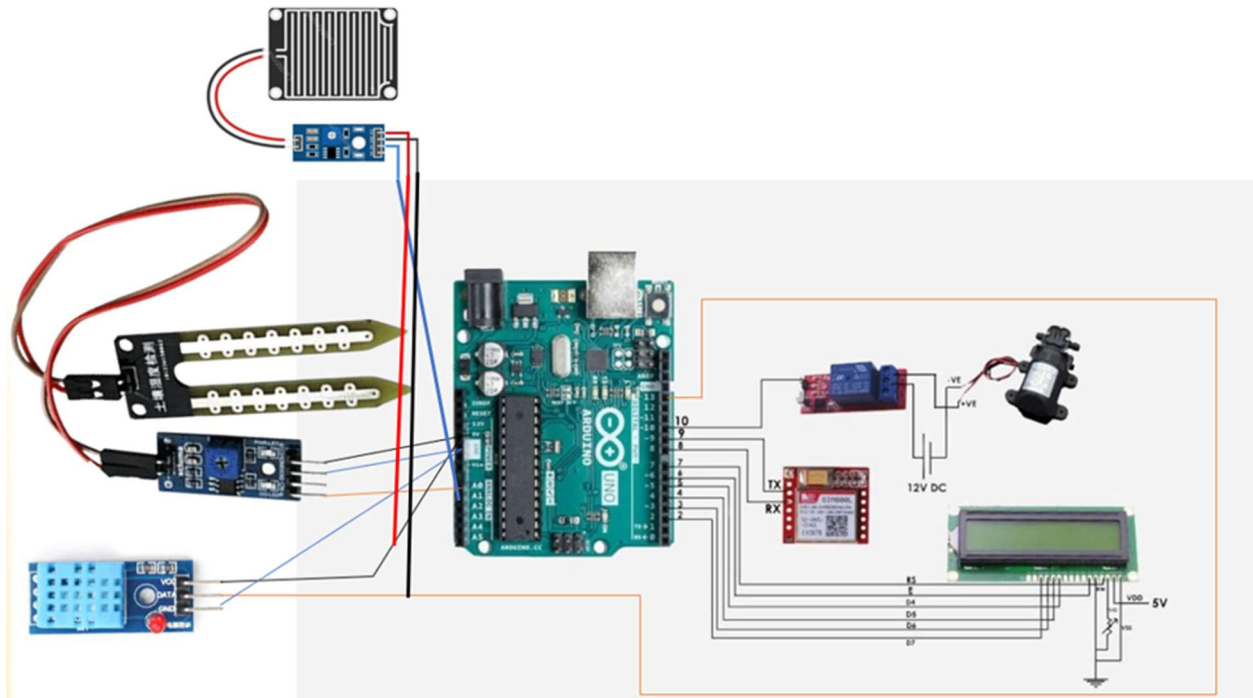
GSM Module Applications:

- The module is the best application to design a graphic for Voice call and SMS application.
- Some IoT applications, mostly in an emergency have the module.
- The location tracing system also uses SIM900A.
- SIM900A can use for mobile communication.

SIM800A GSM Module Main Features

FEATURES	DETAIL
Power Input	3.4V to 4.5V
Operating Frequency	EGSM900 and DCS1800
Transmitting Power Range	2V for EGSM900 and 1W for DCS1800
Data Transfer Link	Download: 85.6kbps, Upload:42.8kbps
SMS	MT, MO, CB, Text and PDU mode.
Antenna Support	Available
Audio Input/output	Available
Serial Port	I2C and UART
Serial Debug Port	Available

6. CIRCUIT DIAGRAM:



7.WORKING:

Initially Working starts from sensing of three parameters from the agricultural field.

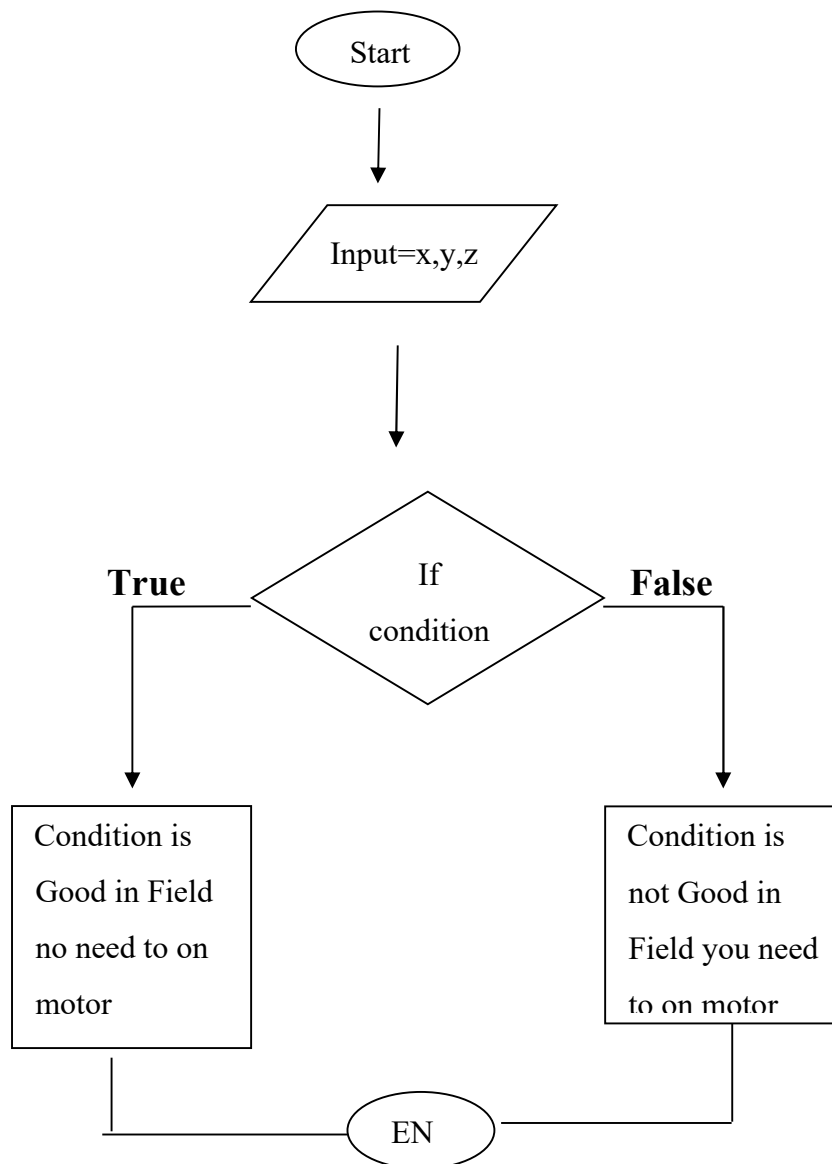
1. Sensing soil moisture content in the soil.
 2. Sensing Humidity and temperature from surroundings.
 3. Sensing the rain fall
- By taking these three as the parameter as inputs to the ARDUINO ,the Transmitter and receiver pins of the ARDUINO will send serial data to the GSM module .
 - When the GSM module receive the data from the ARDUINO,it will automatically send message to the farmer mobile number which was already feeded in the GSM program.
 - The message is about natural conditions in the farm.

Working Flowchart:

H=good

M=medium

L=bad



x=soil moisture

y=temperature and humidity

z=rain

Working (truth table):

Soil moisture	Humidity and Temperature	Rain	GSM Module	Motor on/off
Low	low	low	Send msg	on
low	low	high	Send msg	On/off
low	high	Low	Send msg	On
Low	High	high	Send msg	On/off
High	High	Low	Send msg	Off
High	Low	High	Send msg	Off
high	high	high	Send msg	Off
high	low	Low	Send msg	off

8.Arduino code

```
#include <SoftwareSerial.h>

SoftwareSerial mySerial(9, 10);

#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 17, 2);

#include <dht.h>

#define dht_pin A0

dht DHT;

int sensorPin = A1;

int sensorValue = 0;

float moisturePercentage;

const int sensorMin = 0;

const int sensorMax = 1024;
```

```

int sensorReading =0;

void setup()
{
    mySerial.begin(9600);
    Serial.begin(9600);
    lcd.init();
    lcd.backlight();
    lcd.setCursor(0,0);
    lcd.print("initializing.....");
    delay(1000);
    lcd.setCursor(0,0);
    lcd.print("GSM BASED    ");
    lcd.setCursor(0,1);
    lcd.print("MOTOR CONTROL");
    delay(10000);
    DHT.read11(dht_pin);
    pinMode(dht_pin,INPUT);
    pinMode(sensorPin,INPUT);
}

void loop()
{
    Serial.print("\nGSM BASED MOTOR CONTROL\n");
    lcd.setCursor(0,0);

```

```

lcd.print("Humidity=");

lcd.print(DHT.humidity);

lcd.print("% ");

Serial.print("Humidity=");

Serial.print(DHT.humidity);

Serial.print("%\n");

lcd.setCursor(0,1);

lcd.print("Temp=");

lcd.print(DHT.temperature);

lcd.println("C  ");

Serial.print("Temperature=");

Serial.print(DHT.temperature);

Serial.println("C\n");

delay(1300);

sensorValue = analogRead(sensorPin);

int range = map(sensorValue,sensorMin,sensorMax,0,3);

switch (range)

{

case 0:

lcd.setCursor(0,1);

Serial.print("Moisture=");

lcd.print("Moisture=");

Serial.print(sensorValue);

lcd.print(sensorValue);

lcd.print("    \n");

lcd.setCursor(0,0);

```

```
Serial.println("Over Water");

lcd.println("Over Water    ");

break;

case 1:

lcd.setCursor(0,1);

Serial.print("Moisture=");

lcd.print("Moisture=");

Serial.print(sensorValue);

lcd.print(sensorValue);

lcd.print("    \n");

lcd.setCursor(0,0);

Serial.println("good condition");

lcd.println("good condition  ");

break;

case 2:

lcd.setCursor(0,1);

Serial.print("Moisture=");

lcd.print("Moisture=");

lcd.print(sensorValue);

lcd.print("    \n");

lcd.setCursor(0,0);

Serial.println("Need water");

lcd.println("Need water    ");

break;

}

delay(1500);
```

```

sensorReading = analogRead(A2);

int range1 = map(sensorReading,sensorMin,sensorMax,0,3);

switch (range)
{
case 0:

lcd.setCursor(0,1);

Serial.print("Rain=");

lcd.print("Rain=");

Serial.print(sensorReading);

lcd.print(sensorReading);

lcd.print("    \n");

lcd.setCursor(0,0);

Serial.println("Flood");

lcd.println("Flood    ");

break;

case 1:

lcd.setCursor(0,1);

Serial.print("Rain=");

lcd.print("Rain=");

Serial.print(sensorReading);

lcd.print(sensorReading);

lcd.print("    \n");

lcd.setCursor(0,0);

Serial.println("Rain Warning");

lcd.println("Rain Warning  ");

break;

```

case 2:

lcd.setCursor(0,1);

Serial.print("Rain=");

lcd.print("Rain=");

Serial.print(sensorReading);

lcd.print(sensorReading);

lcd.print(" \n");

lcd.setCursor(0,0);

Serial.println("Not Raining");

lcd.println("Not Raining ");

break;

}

if (Serial.available()>0)

switch(Serial.read())

{

case 's':

SendMessage();

break;

case 'r':

RecieveMessage();

break;

}

if (mySerial.available()>0)

Serial.write(mySerial.read());

}

```

void SendMessage()

{

mySerial.println("AT+CMGF=1"); //Sets the GSM Module in Text Mode

delay(1000); // Delay of 1000 milli seconds or 1 second

mySerial.println("AT+CMGS=\"+917702060720\"\\r"); // Replace x with mobile number

delay(1000);

mySerial.println("INFORMATION FROM GSM BASED MOTOR CONTROL");// The SMS text you
want to send

mySerial.println("TEMPERATURE=");

mySerial.print(DHT.temperature);

mySerial.println("C\\n");

mySerial.println("HUMIDITY=");

mySerial.print(DHT.humidity);

mySerial.println("%\\n");

mySerial.println("MOISTURE=");

mySerial.print(sensorValue);

mySerial.println("\\n");

mySerial.println("RAIN=");

mySerial.print(sensorReading);

mySerial.println("\\n");

mySerial.println("THANK YOU");

delay(100);

mySerial.println((char)26);// ASCII code of CTRL+Z

delay(1000);

}

void RecieveMessage()

```

```
{  
mySerial.println("AT+CNMI=2,2,0,0,0"); // AT Command to receive a live SMS  
delay(1000);  
}
```

9.Applications :

- It is used for effective irrigation of the fields by providing water to the fields automatically according to the information from installed sensors.
- It doesn't need manual check for moisture level in soil.
- It can be used to provide water in nursery planting areas.
- It can also be implemented as Domestic portable water for remote homes, campgrounds etc.

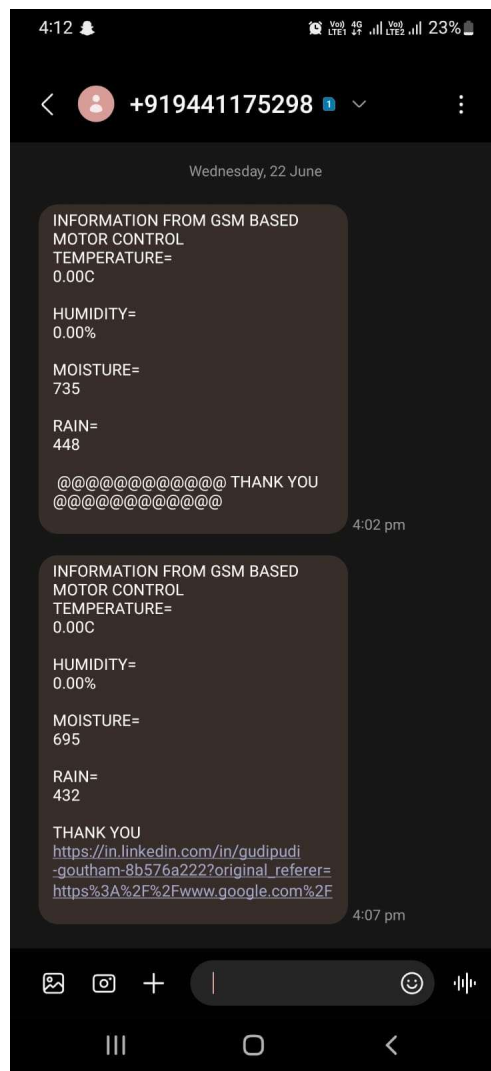
Advantages:

- The proposed system is cost efficient.
- Farmers can invest time in other vital tasks.
- The wastage of water is less and water is saved and productivity is increased.
- Time and labour cost of farmer will also be reduced to great extent.
- Modifications can be made in the circuit anytime depending upon the requirements of the farmer.
- Saves time and reduces manpower.

Disadvantages:

- The farmer must have a phone.
- The software code is programmed, it must be changed for every crop with requirement.
- It would be difficult in case of failure of GSM module.
- The Kit is to be protected from reaching water.
- The kit must be prevented from rainfall too.

10.OUTPUT & Conclusion:



CONCLUSION:

- The need for automation and remote understands ability particularly increase the importance of this system. This system requires only one-time implementation.
- Also, user need not report at the site usually for watering the field. In case of any default such as pump motor is ON or OFF. The user will get report by a normal message.
- Thus, this system ensures the efficient management of water and healthy growth of plants.
- This system is automated irrigation system so it reduces the human resources and also it is feasible and cost effective for optimizing water resources for production.
- We can monitor the status of all the sensors.

References:

- Shen Jin, Song Jingling, Han Qiuyan, Wang Shengde, and YangYan, School of Electric and Electronic Engineering, A Remote Measurement and Control System for Greenhouse Based on GSM-SMS, IEEE 8th International Conference on Electronic Measurement and Instrument, 2007,pp. 45-82
- R.Aarthi and Dr. A. Shaik Abdul Khadir, “An efficient method of irrigation using sensors”, International Journal of advanced research in computer and communication engineering Vol. 4, Issue 7, July 2015.
- A. Padyal, S. Shitole, S. Tilekar, and P. Raut, “Automated Water Irrigation System using Arduino Uno and Raspberry Pi with Android Interface,” Int. Res. J. Eng. Technol., vol. 5, no. 1, pp. 768– 770, 2018.